University of California, Davis
Managerial Economics (ARE) 106, Fall 2025
Instructor: John H. Constantine
Discussion Section 3

Problem 1:
There are three charts with this problem. They are found at the end of problem 1.

(a) Use the data on the chart to give the characteristics of house 1. (Note that there are 1,500 houses in this data set. The "living area" is measured in 100 square feet. "lgelot" states whether the house has a large lot or not. "sprice" is the Stockton, CA house price)
(b) Why is house 2 worth $30,000 less than house 1? Why is house 1 worth a different amount than the other 1,499 houses? What is econometrics attempting to do?
(c) What is the estimated equation? Provide interpretations for b_1 and b_2. State the t values for both b_1 and b_2. Does b_1 make economic sense?
(d) Using $\alpha = 0.05$, and a two-tail test, test the hypothesis that the amount of living area has no impact on price.
(e) Conduct a 95% confidence interval of β_2.
(f) Conduct a 99% confidence interval of β_2.
(g) Do problem (d) again but now using a one-tail test. $\alpha = 0.05$.
(h) Do problem (d) again but now use the p-value. $\alpha = 0.05$.
(i) Test the hypothesis that:

H_0: $\beta_2 = 9{,}000$ vs. H_1: $\beta_2 \neq 9{,}000$

(j) Verify that $R^2 = 0.6287$.
(k) Verify that that standard error $\hat{\sigma}^2$ is 38,556.81.
(l) Use the regression estimates to calculate se(b_2).

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.792876888
R Square	0.62865376
Adjusted R Square	0.628405865
Standard Error	38556.80727
Observations	1500

n

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	3.77004E+12	3.77004E+12	2535.971095	0
Residual	1498	2.22697E+12	1486627387		
Total	1499	5.99701E+12			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 99.0%*	*Upper 99.0%*
Intercept	-30069.19961	3211.568249	-9.362777707	2.74583E-20	-36368.84767	-23769.55154	-38352.20448	-21786.19474
livarea	9181.710844	182.3271998	50.35842625	0	8824.067132	9539.354557	8711.46797	9651.953718

(f) Conduct a 99% confidence interval of β_2.

(g) Do problem (d) again but now using a one tai

Confidence interval

$$\underbrace{\text{estimate of } \beta_2} \pm \underbrace{\left(\text{Standard error of estimate of } \beta_2\right)} * \underbrace{t_{0.01}}$$

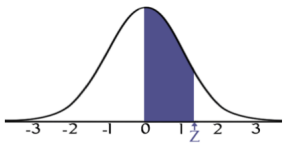
$$9181.71 \pm (182.33) * 2.55$$

any number in this range is a confidence interval

in

sample size

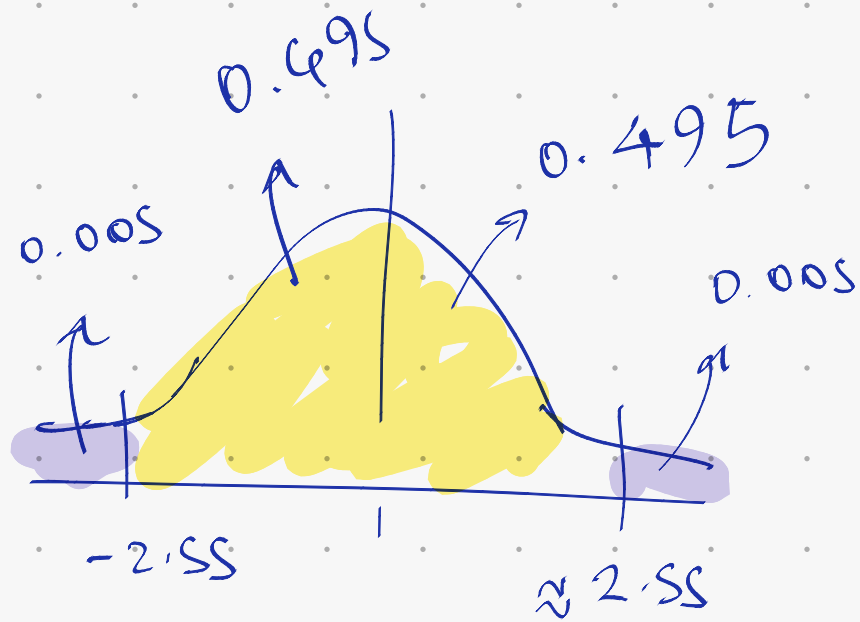
as $n = 1500$, we use z-table (basically degree of freedom is ∞ (> 50)).



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998



we use $t_{0.01} = 2.55$

(1.65, 1.46, 2.55)

10% 5% 1%

(d) Using $\alpha = 0.05$, and a two-tail test, test the hypothesis that the amount of living area has no impact on price.

(e) Conduct a 95% confidence interval of β_2.

(f) Conduct a 99% confidence interval of β_2.

(g) Do problem (d) again but now using a one-tail test. $\alpha = 0.05$.

(right-tail test, if effect is positive)

① we build the hypothesis. $\underbrace{H_0: \beta_2 \leq 0, \quad H_1: \beta_2 > 0}_{\text{right-tail (one-tail) test}}$

② The Rule for right-tail test is

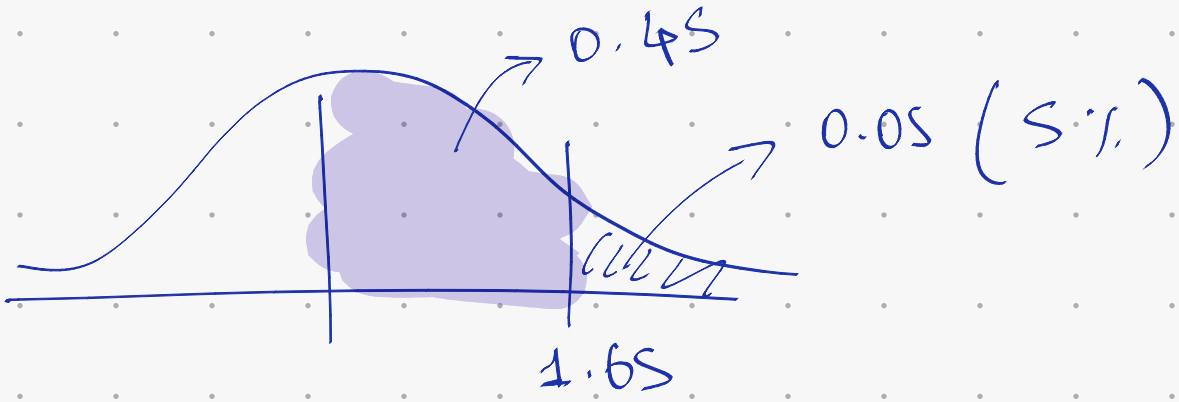
Reject if $t\text{-stat} > t_\alpha$ ($t_{critical}$)

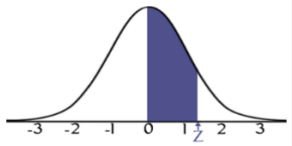
Rule: Reject if t-stat > t_α

$$t\text{-stat} = \frac{\text{estimate of } \beta_2 - \text{null value}}{\text{standard error of estimate of } \beta_2}$$

$$= \frac{9181.71 - 0}{182.32} = 50.36$$

t_α is





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0.1	0.0398	0.0438	0.0478
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2.0	0.4772	0.4778	0.4783
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2.2	0.4861	0.4864	0.4868
2.3	0.4893	0.4896	0.4898
2.4	0.4918	0.4920	0.4922
2.5	0.4938	0.4940	0.4941
2.6	0.4953	0.4955	0.4956
2.7	0.4965	0.4966	0.4967
2.8	0.4974	0.4975	0.4976
2.9	0.4981	0.4982	0.4982
3.0	0.4987	0.4987	0.4987
3.1	0.4990	0.4991	0.4991
3.2	0.4993	0.4993	0.4994
3.3	0.4995	0.4995	0.4995
3.4	0.4997	0.4997	0.4997

<u>Final step</u> As $50.36 > 1.65$ we reject the hypothesis that $\beta_2 \leq 0$.

(h) Do problem (d) again but now use the p-value. $\alpha = 0.05$.

Rule: Reject if p-val $< \alpha$.

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 99.0%	Upper 99.0%
Intercept	-30069.19961	3211.568249	-9.362777707	2.74583E-20	-36368.84767	-23769.55154	-38352.20448	-21786.19474
livarea	9181.710844	182.3271998	50.35842625	0	8824.067132	9539.354557	8711.46797	9651.953718

final step As 0 is smaller than 0.05 we reject the null Hypothesis that $\beta_2 = 0$.

Definition of p-value:

(i) Test the hypothesis that:

$H_0: \beta_2 = 9{,}000$ vs. $H_1: \beta_2 \neq 9{,}000$

Rule

I do a two-tail test at $\alpha = 0.05$. This means that I reject the null hypothesis if my t-stat > 1.96.

$$t\text{-stat} = \frac{b_2 - \text{null value}}{se(b_2)}$$

$$= \frac{9181.71 - 9000}{182.3272} = 0.997$$

<u>final step</u> As $0.997 \not> 1.96$, I fail to reject the null that $\beta_2 = 9000$.